OSS Audit of VLC Media Player

Student Name: Mishita Tiwari

Registration Number : 24BAI10204

Course: B.Tech CSE (AI/ML), VIT Bhopal

Abstract :-

The aim of this project is to perform an Open Source Software (OSS) audit on VLC Media Player. The audit involves checking system information, installed packages, disk permissions, log files, and creating a small OSS manifesto. The project demonstrates how Linux commands and scripts can help analyze OSS projects, even on a Windows system using PowerShell or Git Bash.

Objective :-

Understand OSS audit techniques - Practice writing bash scripts for system analysis - Document system outputs and security insights - Create a small manifesto for OSS understanding

Project Description :-

The project consists of 5 scripts, each performing a separate audit task: 1. **system_info.sh** – Collects system details: username, OS, kernel, date, and time. 2. **package_inspector.sh** – Checks for the presence of VLC Media Player (simulated installation check). 3. **disk_perm_audit.sh** – Displays disk usage and folder permissions to analyze access rights. 4. **log_analyze.sh** – Searches system log files for errors or warnings (simulated). 5. **manifesto.sh** – Generates a small OSS manifesto based on user input.

All scripts were executed successfully, and their outputs were captured in screenshots.

Folder Structure

```
linux_project/
├── system_info.sh
├── package_inspector.sh
├── disk_perm_audit.sh
├── log_analyze.sh
├── manifesto.sh
├── report.docx
```

```
└── screenshots/
    ├── system_info.png
    ├── package_inspector.png
    ├── disk_perm_audit.png
    ├── log_analyze.png
    └── manifesto.png
```

Scripts Explanation

1. **system_info.sh** - Displays current username, OS info, kernel version, date, and time. - Commands used: echo, uname, date.

```
Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ cd "C:\Users\Mishita Tiwari\Desktop\Linux Project"

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ ls
disk_perm_audit.sh*  log_analyze.sh*  manifesto.sh*  package_inspector.sh*  system_info.sh*

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ chmod +x system_info.sh
./system_info.sh
==== System Identity Report ====
Kernel: 3.6.6-1cdd4371.x86_64
User: Mishita Tiwari
Date/Time: Mon, Mar 30, 2026  2:10:03 AM
Linux Distro: Simulated Ubuntu
Open Source Focus: This system uses open source tools.
```

2. **package_inspector.sh** - Checks if VLC Media Player is installed (simulated using conditional statements). - Commands used: dpkg -l | grep vlc (or PowerShell equivalent).

```
Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ chmod +x package_inspector.sh

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ ./package_inspector.sh
Checking package: vlc (simulation)
vlc is NOT installed (simulated for Windows)
```

3. **disk_perm_audit.sh** - Checks disk usage and permissions of folders/files. - Commands used: ls -l, df -h.

```
Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ chmod +x disk_perm_audit.sh

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ ./disk_perm_audit.sh
Disk usage and folder permissions (simulated)
/etc: drwxr-xr-x
/var/log: drwxr-xr-x
/c/Users/Mishita Tiwari: drwx------
```

4. **log_analyze.sh** - Searches log files for errors and warnings. - Commands used: grep “error” /var/log/* (simulated for Windows).

```
Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ chmod +x log_analyze.sh

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ ./log_analyze.sh
Analyzing logs (simulation)
Found 0 lines with keyword 'error' in /var/log/syslog (simulated for Windows)
```

5. **manifesto.sh** - Prompts user for OSS inputs and prints a small OSS manifesto. - Demonstrates script interaction and file writing.

```
Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ chmod +x manifesto.sh

Mishita Tiwari@LAPTOP-KCITACB2 MINGW64 ~/Desktop/Linux Project (master)
$ ./manifesto.sh
Enter one open source tool you use daily:
VLC
Enter one word for freedom:
Freedom
Generated OSS Manifesto:
I choose VLC because it represents Freedom in the digital world.
```

Conclusion :-

This project provided hands-on experience in: - Writing bash scripts for system auditing - Understanding OSS audit techniques - Capturing and documenting outputs for project submission - Simulating Linux behavior on a Windows system

All objectives were successfully achieved, and the project demonstrates a basic OSS audit workflow suitable for academic purposes.